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High-frequency monitoring of growth at risk[☆]

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ABSTRACT

Monitoring changes in financial conditions provides valuable information on the contribution of financial risks to future economic growth. For that purpose, central banks need real-time indicators to promptly adjust their policy stance. In this paper, we extend the quarterly growth-at-risk (GaR) approach of Adrian et al. (2019) by accounting for the high-frequency nature of financial conditions indicators. Specifically, we use Bayesian mixed-data sampling (MIDAS) quantile regressions to exploit the information content of both a financial stress index and a financial conditions index, leading to real-time high-frequency GaR measures for the euro area. We show that our daily GaR indicator (i) displays good GDP nowcasting properties, (ii) can provide an early signal of GDP downturns, and (iii) allows day-to-day assessment of the effects of monetary policies. During the first six months of the Covid-19 pandemic period, it has provided a timely measure of the tail risks to euro-area GDP.

1. Introduction

There is a growing body of research, both theoretical and empirical, suggesting that the information content of financial indicators is relevant for macroeconomic forecasting and that financial shocks are influential in driving global activity.[1] It follows that a good understanding of the evolution of economic conditions requires monitoring tension in the financial sector. Based on this observation, Adrian et al. (2019) developed a tool for evaluating financial risks to economic growth, using a tail-risk approach known as growth at risk (GaR), that can be seen as equivalent to the value-at-risk concept in finance. This approach is used to keep track of the distortion of the entire expected growth distribution according to financial market developments using quantile regression methods (Giglio et al., 2016). Quantile regressions provide an estimate of the elasticity of the gross domestic product (GDP) growth rate to financial conditions for any range of values of the economic growth rate, thus capturing the non-linear nature of this relationship. Fig. 1 displays the values of these elasticities for various quantiles in the United States and the euro area and shows that the

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1 A few prominent contributions are Nolan and Thoenissen (2009), Helbling et al. (2011), Claessens et al. (2012), Jermann and Quadrini (2012), Gilchrist and Zakrajsek (2012), Christiano et al. (2014), and Lopez-Salido et al. (2017).

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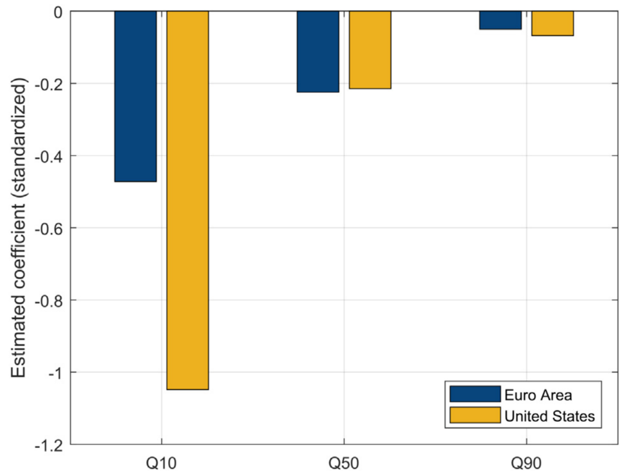

Fig. 1. Elasticity of GDP growth to quarterly financial conditions for different quantiles. *Notes:* Q10, Q50, and Q90 represent the 10% quantile, 50% quantile (or median), and 90% quantile, respectively. Estimates are obtained by replicating the one-step-ahead quarterly model in Adrian et al. (2019) over the 2001Q1–2019Q4 sample. The euro-area financial conditions index is the one proposed by Petronevich and Sahuc (2019) and the U.S. financial conditions index comes from the Chicago Fed (NFCI).

relationship is strongest for changes in GDP located at the bottom of the distribution. This illustrates that a tightening of financial conditions tends to amplify the effects of negative shocks to the real economy, as notably emphasized by Bernanke and Gertler (1989), while an easing of these conditions has a more limited impact on economic activity at the peak of the cycle.

However, the standard GaR approach suffers from several drawbacks arising primarily from modeling tail risks based on quarterly data, while financial indicators are often sampled at a higher frequency. To ensure the same frequency, financial conditions indexes are usually aggregated by simple averaging to get the data sampled at the same low frequency as GDP. Such data aggregation is likely to lead to biased estimates if the underlying data generating process does not feature a flat aggregation scheme from high to low frequencies. Asymptotically inefficient and inconsistent estimates may hence dampen the information content of daily financial indicators and have adverse effects on forecasting. In addition, this strategy makes the standard GaR rely on somewhat outdated financial information, the latter usually entering the model with lag of one quarter, while central banks need the most current information to adjust their policy stance promptly. Higher-frequency data series are thus helpful to more accurately assess economic conditions, as evidenced by Clements and Galvao (2009), Rodriguez and Puggioni (2010), Schorfheide and Song (2015), Ghysels (2016), Mazzi and Mitchell (2020), Lima et al. (2020), and Carriero et al. (2020). Particularly, Andreou et al. (2013) show that model regressions exploiting high-frequency financial data series provide a substantial gain regarding current and next-quarter output growth forecasting accuracy against various benchmark models and survey forecasts.

In this paper, we implement a mixed-data sampling (MIDAS) quantile regression approach (Ghysels et al., 2016) to obtain a real-time high-frequency GaR measure. Specifically, we consider a Bayesian approach for estimation (Korobilis, 2017; Carriero et al., 2020) and a restricted Almon lag polynomial approximation of the high-frequency component (Mogliani & Simoni, 2021).[2] We then consider the 10th percent quantile of the conditional predictive distribution of current euro-area GDP growth—which is akin to nowcasting—and compute a high-frequency measure of current tail risks on activity that we call the *daily* GaR(10%). Our model combines information stemming from two daily euro-area financial indicators in order to better capture different features of the financial side of the economy: (i) the Composite Indicator of Systemic Stress (CISS) of the European Central Bank (Holló et al., 2012), and (ii) the financial indicator proposed by Petronevich and Sahuc (2019). The first one is a financial stress indicator designed to react more to systemic fragility in financial markets, whereas the second one is a financial conditions index, which is more useful for exploring macro-financial linkages. We also collect seasonally and calendar-adjusted vintages of quarterly GDP into a real-time triangle spanning from 1999Q1 to 2020Q2 to perform a historical analysis on a pseudo-real-time basis.

We propose several applications to highlight the practical interest of our daily GaR(10%) measure. We first evaluate the nowcasting ability of our model, i.e. the ability to assess current GDP growth, based on the entire predictive distribution. Second, we look at the real-time evolution of the indicator before and during a specific recession episode, namely the sovereign debt crisis that affected the euro area from 2010 to 2013. Third, we inspect the link between the GaR measure and the main unconventional monetary policy decisions announced by the European Central Bank between 2013 and 2018. Finally, we focus on the Covid-19 pandemic period during the first half of 2020, which offers an interesting case

[2] Carriero et al. (2020) showed that Bayesian mixed-frequency quantile regressions may provide superior predictive accuracy compared to the frequentist approach.

study for assessing extreme macroeconomic risks through our high-frequency measure.

We show that our high-frequency approach provides an efficient way to monitor financial risks weighing on the euro area. For instance, our daily GaR(10%) measure would have led to an advanced detection of the GDP downturn observed during the European sovereign debt crisis, by steadily declining by approximately one percentage point, more than a quarter ahead of the start of the recession in 2011Q4. In addition, it provides a day-to-day benchmark for monetary policy by revealing information about how economic activity is likely to react to new announcements. We observe in particular an increase in the value of the GaR(10%) around (either following or anticipating, consistently with the narrative and the expectations prevailing at that time) each announcement of the main unconventional monetary policies between 2013 and 2018. Finally, during the first half of 2020, it provided a timely indication of tail risks on euro-area GDP, especially since the World Health Organization (WHO) announcement recognizing Covid-19 as a global pandemic on March 11th, 2020.

Our paper contributes to very recent literature on the use of quantile regressions to evaluate macroeconomic risks (see Giglio et al., 2016, Adrian et al., 2019, Gonzalez-Rivera et al., 2019, Figueres & Jarocinski, 2020, and Adams et al., 2021), as well as on mixed-frequency data models to assess current economic conditions. In addition, we do not solely focus on nowcasting but also propose a new piece to the policymakers' toolkit for real-time macro-financial surveillance.

In the remainder of the paper, Section 2 introduces the growth-at-risk approach with mixed-data sampling and describes the Bayesian approach. Section 3 presents the data and the computational approach. Section 4 proposes some applications, including a focus on the Covid-19 crisis, and Section 5 concludes.

2. Growth at risk with mixed-frequency data sampling

2.1. The growth-at-risk approach

Since the Global Financial Crisis and the ensuing Great Recession in 2008–09, financial institutions have had to step up their monitoring of financial conditions in order to be able to react promptly to any possible financial shocks before their transmission to the real economy. Adrian et al. (2019) recently developed a methodology, referred to as growth at risk (GaR), for measuring financial risks or vulnerabilities to U.S. economic growth. This approach is now widely used by the International Monetary Fund to assess risks to the global financial system in its flagship biannual *Global Financial Stability Report* (see, for instance, IMF, 2019).

The GaR approach relies on a quantile regression of GDP growth on past financial conditions and past GDP, accounting thus for non-linearities in a very simple econometric model of macro-financial linkages. Indeed, both theoretical and empirical studies have shown that financial markets play a key role in the transmission and propagation of shocks to the economy, but the channels of transmission are highly complex and present a strong degree of non-linearity. For instance, building on earlier theoretical contributions such as Bernanke and Gertler (1989), Carlstrom and Fuerst (1997), Kiyotaki and Moore (1997), and Bernanke et al. (1999), the recent literature shows that financial constraints can lead to highly non-linear dynamics in the economy's response to shocks (asymmetric impulse responses following a negative or a positive shock). Recently, Hubrich and Tetlow (2015) empirically assessed models of financial friction and showed that (i) a single-regime model of the macroeconomy and financial stress is inadequate to capture the dynamics of the economy, and (ii) output reacts more strongly to financial shocks in times of financial stress than in normal times.

As in Adrian et al. (2019), let us assume that we want to assess the joint effect of past GDP growth (y_{t-h}) and a given financial conditions indicator (x_{t-h}), where h is the forecast horizon, on current GDP growth (y_t). Furthermore, consider an additional vector $\mathbf{w}_{t-h}$ of K control variables. We assume at this stage that both the target variable and the vector of regressors have been sampled at the same quarterly frequency. The methodology for dealing with frequency mismatches is the core of this paper and will be presented in the following section. Beyond the standard linear ordinary least squares (OLS) approach, the quantile regression framework put forward by Koenker and Bassett (1978) is an efficient way to introduce non-linearities in the relationship between a given variable y_t and its predictors. Instead of minimizing the sum of squared errors, as in the OLS approach, quantile estimation is based on the asymmetric minimization of the weighted absolute errors.

Let us consider the following quantile regression:

$$y_t = \beta_1(\tau)y_{t-h} + \beta_2(\tau)x_{t-h} + \boldsymbol{\gamma}(\tau)'\mathbf{w}_{t-h} + \epsilon_t, \tag{1}$$

where the vector of coefficients $\boldsymbol{\beta}(\tau) := (\beta_1(\tau), \beta_2(\tau), \boldsymbol{\gamma}(\tau)')'$ depends on the τ-th quantile of the random error term ϵ_t. The coefficients $\widehat{\boldsymbol{\beta}}(\tau)$ are obtained by minimizing the following loss function:

$$\sum_{t=1}^{T} \rho_\tau \left(y_t - \boldsymbol{\beta}(\tau)'\mathbf{z}_{t-h}\right), \tag{2}$$

where $\mathbf{z}_{t-h} = (y_{t-h}, x_{t-h}, \mathbf{w}_{t-h})'$, $\rho_\tau(u) = u(\tau - I(u < 0))$ is the check loss function, with $I(\cdot)$ denoting the indicator function. Koenker and Bassett (1978) proved that the predicted value $\widehat{Q}_{y_t}(\tau|\mathbf{z}) = \widehat{\boldsymbol{\beta}}(\tau)'\mathbf{z}_{t-h}$ is a consistent linear estimator of the conditional quantile function of y_t. In order to provide an evaluation of financial risks to future economic activity, an estimate of the future quantile function of $y_{T|T-h}$, conditional on sample information available up to $T-h$, is given by:

$$\widehat{Q}_{y_{T|T-h}}(\tau|\mathbf{z}) = \widehat{\boldsymbol{\beta}}(\tau)'\mathbf{z}_{T-h}. \tag{3}$$

Based on estimates of the conditional quantile function over a discrete number of quantile levels, it is possible to estimate the full continuous conditional distribution of $y_{T|T-h}$. As in Adrian et al. (2019), we choose to fit a flexible distribution, known as the generalized skewed Student's distribution, in order to smooth the estimated conditional quantile function of $y_{T|T-h}$ and recover a probability density function.[3] This specific distribution allows for fat tails and asymmetry and boils down to a normal distribution as a specific case. The generalized skewed Student's distribution has the following density function:

$$f(y;\mu,\sigma,\alpha,\nu) = \frac{2}{\sigma}\, t\left(\frac{y-\mu}{\sigma};\nu\right) T\left(\alpha\frac{y-\mu}{\sigma}\sqrt{\frac{\nu+1}{\nu+\left(\frac{y-\mu}{\sigma}\right)^2}};\nu+1\right), \tag{4}$$

where μ is a location parameter, σ is a scale parameter, ν is a fatness parameter, α is a shape parameter, and $t(\cdot)$ and $T(\cdot)$ respectively denote the probability density function and the cumulative density function of the standard Student's distribution (Azzalini & Capitanio, 2003).

In practice, the four parameters of the generalized skewed Student's distribution are estimated through a quantile matching approach aiming at minimizing the squared distance between the estimated conditional quantile functions and the inverse cumulative density function of the generalized skewed Student's distribution given by:

$$\min_{\mu,\sigma,\alpha,\nu}\sum_{\tau}\left[\widehat{Q}_{y_{T|T-h}}(\tau|\mathbf{z}) - F^{-1}(\tau;\mu,\sigma,\alpha,\nu)\right]^2, \tag{5}$$

where $F^{-1}(\cdot)$ is the inverse cumulative skewed Student's distribution. Finally, from the fitted quantile function, $F^{-1}(\tau;\widehat{\mu},\widehat{\sigma},\widehat{\alpha},\widehat{\nu})$, it is possible to compute some downside risk measures, such as the expected shortfall, at any given probability level. Due to the short data sample used in our empirical part, we focus on the lower 10th percent quantile of the predicted distribution (see also Figueres & Jarocinski, 2020), called the GaR(10%), which is given by:

$$Q^*_{y_{T|T-h}}(\tau=0.10|\mathbf{z}) := F^{-1}(\tau=0.10;\widehat{\mu},\widehat{\sigma},\widehat{\alpha},\widehat{\nu}). \tag{6}$$

This can be interpreted as the expected value of future GDP at 10% probability, stemming from the conditional quantile function of $y_{T|T-h}$.

2.2. Introducing MIDAS quantile regression

The problem with the setup described above is that both the aggregation of high-frequency (financial) indicators into the lower frequency of GDP and the lag structure of the specification in Eq. (1) prevent the model from reacting readily to sudden shocks. Hence, from the point of view of policymakers, the GaR model appears to be an impractical tool for monitoring financial risks to activity in real time.

We thus propose to adapt Eq. (1) to account for the possible high-frequency nature of the regressors. Let us assume that the financial indicator x_t is available on a daily basis, i.e. virtually without delay, and denote it $x_t^{(d)}$ (i.e. it is observed about $d = 60$ times on average between quarters $t-1$ and t). Further, let us note that the set of control variables can also be sampled at a higher frequency than the target variable. Hence, for notational convenience and consistently with the applications presented in the next sections, let us assume that $K = 1$ and that w_t is sampled at a monthly frequency, and let us denote it by $w_t^{(m)}$, with $m = 3$. According to these features, we can build a high-frequency real-time GaR measure which relates current quarterly GDP growth to past *and* current values (up to the latest available observation) of high-frequency financial conditions and the control variable. For this purpose, the general model used throughout this paper follows mixed-data sampling (MIDAS) quantile regression (MIDAS-QR):

$$y_t = \beta_1(\tau)y_{t-1} + \beta_2(\tau)\sum_{c=0}^{C_x-1}\widetilde{B}\left(c;\boldsymbol{\theta}_x(\tau)\right)L^{c/d}x^{(d)}_{t-h_d} + \gamma(\tau)\sum_{c=0}^{C_w-1}\widetilde{B}\left(c;\boldsymbol{\theta}_w(\tau)\right)L^{c/m}w^{(m)}_{t-h_m} + \epsilon_t, \tag{7}$$

where $\widetilde{B}\left(c;\boldsymbol{\theta}_j(\tau)\right)$ is a weighting function (normalized to sum up to 1), which depends on a vector of parameters $\boldsymbol{\theta}_j(\tau)$, for $j = \{x, w\}$, and a lag order c. Note that the forecast horizon is expressed in high-frequency terms ($h_f = 0, 1/f, 2/f, \ldots, (C_j-1)/f$, for $f = \{d, m\}$). While Ghysels et al. (2016) propose the Beta lag polynomial function for the quantile weighting function, we choose a simple polynomial approximation of the underlying true weighting structure provided by the (un-normalized) Almon lag polynomial $B\left(c;\boldsymbol{\theta}_j(\tau)\right) = \sum_{i=0}^{p}\theta_{i,j}(\tau)c^i$, where $\boldsymbol{\theta}_j(\tau) := \left(\theta_{0,j}(\tau), \theta_{1,j}(\tau), \ldots, \theta_{p,j}(\tau)\right)'$, similarly to Lima et al. (2020) and Mogliani and Simoni (2021). Under the so-called "direct method", Eq. (7) with (un-normalized) Almon lag polynomials can be reparameterized as:

$$y_t = \beta_1(\tau)y_{t-1} + \boldsymbol{\theta}_x(\tau)'\widetilde{\mathbf{x}}^{(d)}_{t-h_d} + \boldsymbol{\theta}_w(\tau)'\widetilde{\mathbf{w}}^{(m)}_{t-h_m} + \epsilon_t, \tag{8}$$

where $\boldsymbol{\theta}_x(\tau)$ and $\boldsymbol{\theta}_w(\tau)$ are vectors featuring $(p+1)$ parameters, $\widetilde{\mathbf{x}}^{(d)}_t := \mathbf{Q}_x\mathbf{x}^{(d)}_t$ and $\widetilde{\mathbf{w}}^{(m)}_t := \mathbf{Q}_w\mathbf{w}^{(m)}_t$ are $((p+1)\times 1)$ vectors of linear combinations of the observed high-frequency variables, $\mathbf{x}^{(d)}_t := \left(x^{(d)}_t, x^{(d)}_{t-1/d}, \ldots, x^{(d)}_{t-(C_x-1)/d}\right)'$ and $\mathbf{w}^{(m)}_t := \left(w^{(m)}_t, w^{(m)}_{t-1/m}, \ldots, w^{(m)}_{t-(C_w-1)/m}\right)'$ are $(C_j\times 1)$ are vectors of high-frequency lags, and $\mathbf{Q}_j$ is a $(p+1\times C_j)$ polynomial weighting matrix (for $j = \{x, w\}$), with $(i+1)$-th row $[0^i, 1^i, 2^i, \ldots, (C_j-1)^i]$ for $i = 0, \ldots, p$. Note that estimates of the slope coefficients $\beta_2(\tau)$ and $\gamma(\tau)$ in Eq. (7) can be computed as $\widehat{\beta}_2(\tau) = \widehat{\boldsymbol{\theta}}_x(\tau)'\mathbf{Q}_x\boldsymbol{\iota}_{C_x}$ and $\widehat{\gamma}(\tau) = \widehat{\boldsymbol{\theta}}_w(\tau)'\mathbf{Q}_w\boldsymbol{\iota}_{C_w}$, where $\boldsymbol{\iota}_{C_j}$ is a $(C_j\times 1)$ vector of ones.

The main advantage of the Almon lag polynomial is that Eq. (8) is linear and parsimonious, as it depends only on $(p+1)$ parameters for the high-frequency variable. Further, linear restrictions on the value and slope of the lag polynomial $B\left(c;\boldsymbol{\theta}_j(\tau)\right)$ may be placed for any $c\in(0, C_j-1)$. Endpoint restrictions, such as $B\left(C_j-1;\boldsymbol{\theta}_j(\tau)\right) = 0$ and $\nabla_c B\left(c;\boldsymbol{\theta}_j(\tau)\right)|_{c=C_j-1} = 0$, may be desirable and economically meaningful, as they jointly constrain the weighting structure to tail off slowly to zero (Mogliani & Simoni, 2021). As a result, the number of parameters in Eq. (8) reduces from $(p+1)$ to $(p-r+1)$, where $r\le p$ is the number of restrictions.

[3] As noted by Adrian et al. (2019), Eq. (3) represents an approximate estimate of the quantile function, which is difficult to map into a probability distribution function, due to approximation error and estimation noise.

2.3. Bayesian estimation

In standard quantile regression (Koenker & Bassett, 1978), the distribution of the residuals ϵ_t in Eq. (8) is unspecified (a non-parametric distribution) and the estimation of the τ-th quantile regression coefficients is the solution to the minimization of the loss function given by (2).

Let us denote $\mathbf{X}_t = (y_{t-1}, \widetilde{\mathbf{x}}^{(d)}_{t-h_d}, \widetilde{\mathbf{w}}^{(m)}_{T-h_m})'$ and $\boldsymbol{\Theta}(\tau) = (\beta_1(\tau), \boldsymbol{\theta}_x(\tau)', \boldsymbol{\theta}_w(\tau)')'$. Yu and Moyeed (2001) showed that the minimization of $\sum_{t=1}^{T} \rho_\tau \left(y_t - \boldsymbol{\Theta}(\tau)'\mathbf{X}_t\right)$ is equivalent to maximizing a likelihood function under the asymmetric Laplace error distribution (ALD) for ϵ_t.[4]

According to Kozumi and Kobayashi (2011), the ALD $f(\epsilon|\sigma)$ can be viewed as a mixture of an exponential and a scaled normal distribution. Considering the random variables $\nu \sim \text{Exp}(1)$ and $\omega \sim \mathcal{N}(0, 1)$, then $\epsilon = \xi_1\sigma\nu + \xi_2\sigma\sqrt{\nu}\omega$ follows the skewed distribution $f(\epsilon|\sigma)$ above, with:

$$\xi_1 = \frac{1-2\tau}{\tau(1-\tau)} \quad \text{and} \quad \xi_2^2 = \frac{2}{\tau(1-\tau)}.$$

Hence, using this expression for ϵ, we can rewrite Eq. (8) as:

$$y_t = \boldsymbol{\Theta}(\tau)'\mathbf{X}_t + \xi_1\tilde{\nu}_t + \xi_2\sqrt{\sigma\tilde{\nu}_t}\omega_t, \tag{9}$$

where $\tilde{\nu}_t = \sigma\nu_t$ follows the exponential distribution $\text{Exp}(\sigma)$, with density function $f(\tilde{\nu}_t|\sigma) = \sigma^{-1}\exp(-\tilde{\nu}_t/\sigma)$. Then, the conditional likelihood function stems from a normal distribution and takes the following form:

$$f(y|\mathbf{X}, \boldsymbol{\Theta}, \tilde{\nu}, \sigma, \tau) \propto \prod_{t=1}^{T} \frac{1}{\xi_2\sqrt{\sigma\tilde{\nu}_t}} \exp\left[-\frac{1}{2}\sum_{t=1}^{T} \frac{\left(y_t - \boldsymbol{\Theta}(\tau)'\mathbf{X}_t - \xi_1\tilde{\nu}_t\right)^2}{\xi_2^2\sigma\tilde{\nu}_t}\right].$$

We consider a standard conditionally normal prior that leads to the following hierarchical representation of the Bayesian MIDAS quantile regression (BMIDAS-QR):

$$\begin{aligned}
y|\mathbf{X}, \boldsymbol{\Theta}, \sigma, \tilde{\boldsymbol{\nu}}, \tau &\sim \mathcal{N}\left(\boldsymbol{\Theta}(\tau)'\mathbf{X}_t + \xi_1\tilde{\nu}_t, \xi_2^2\sigma\tilde{\boldsymbol{\nu}}\right), \\
\boldsymbol{\Theta}|\tau &\sim \mathcal{N}(\boldsymbol{\Theta}_0, \Sigma_0), \\
\tilde{\nu}|\sigma &\sim \text{Exp}(\sigma), \\
\sigma &\sim \text{Inv-Gamma}(a_1, b_1).
\end{aligned}$$

As shown notably by Khare and Hobert (2012), the full conditional posteriors are given by:

$$\begin{aligned}
&\boldsymbol{\Theta}|\mathbf{X}, \sigma, \tilde{\boldsymbol{\nu}}, \tau \sim \mathcal{N}\left(\mathbf{A}^{-1}\mathbf{B}, \mathbf{A}^{-1}\right), \\
&\tilde{\nu}_t|\mathbf{X}, \boldsymbol{\Theta}, \sigma, \tau \sim \text{GiG}\left(\frac{1}{2}, \frac{(y_t - \boldsymbol{\Theta}(\tau)'\mathbf{X}_t)}{\xi_2^2\sigma\tilde{\nu}_t}, \frac{\xi_1^2 + 2\xi_2^2}{\sigma\xi_2^2}\tilde{\nu}_t\right), \\
&\sigma|\mathbf{X}, \boldsymbol{\Theta}, \tilde{\boldsymbol{\nu}}, \tau \sim \text{inv-Gamma}\left(\frac{3T}{2} + a_1, \sum_{t=1}^{T}\frac{(y_t - \boldsymbol{\Theta}(\tau)'\mathbf{X}_t - \xi_1\tilde{\nu}_t)^2}{2\xi_2^2\tilde{\nu}_t} + \sum_{t=1}^{T}\tilde{\nu}_t + b_1\right),
\end{aligned}$$

where $\mathbf{A} = \left(\mathbf{X}_t'\mathbf{D}^{-1}\mathbf{X}_t + \Sigma_0^{-1}\right)^{-1}$, $\mathbf{B} = \mathbf{X}_t'\mathbf{D}^{-1}(y_t - \xi_1\tilde{\nu}) + \Sigma_0^{-1}\boldsymbol{\Theta}_0$, and $\mathbf{D} = \text{diag}(\xi_2^2\sigma\tilde{\nu}_t)$.

3. Data and computational approach

3.1. Data

In our applications, we focus on recent events that have impacted the euro-area economy. As an aggregate measure of economic activity, we use the quarter-on-quarter growth rate of GDP for the euro area as a whole (Fig. 2(a)). To perform the analysis on a pseudo-real-time basis, we collect seasonally and calendar-adjusted vintages of quarterly GDP from Eurostat and the ECB. The data are composed of multiple releases for the same vintage (preliminary flash estimates, flash estimates, and regular estimates), whose number and publication delays vary over time and are collected into a real-time triangle spanning from 1999Q1 to 2019Q4. Actual historical release dates are also identified and matched with each vintage.

As regards the high-frequency financial indicators, we consider two alternative daily euro-area time series: *(i)* a financial stress indicator, which is designed to react to systemic fragilities within financial markets, and *(ii)* a financial conditions index, which is more useful in exploring macro-financial linkages. The two indicators complement each other in capturing different features of the financial side of the economy, as can be seen in Fig. 2(b).[5]

The financial stress indicator is the Composite Indicator of Systemic Stress (CISS) developed by the European Central Bank (Holló et al., 2012). The main methodological innovation of the CISS is the application of basic portfolio theory to the aggregation of five market-specific sub-indexes, namely the foreign exchange market, the equity market, the money market, the bond market, and the financial intermediaries. The aggregation takes into account time-varying cross-correlations between the five sub-indexes. As a result, the CISS puts relatively more weight on situations in which stress prevails in several market segments at the same time, capturing the idea that financial stress is more systemic and thus more dangerous for the economy as a whole if financial instability spreads more widely across the whole financial system. Holló et al. (2012) proposed the determination of critical levels for the CISS using the endogenous outcomes of two econometric regime-switching models.

The financial conditions index (FCI) is the one proposed by Petronevich and Sahuc (2019). This new FCI is

[4] The ALD has density

$$f(\epsilon|\sigma) = \frac{\tau(1-\tau)}{\sigma}\exp\left[-\frac{\rho_\tau(\epsilon)}{\sigma}\right],$$

and moments

$$E(\epsilon) = \sigma\frac{1-2\tau}{\tau(1-\tau)} \qquad V(\epsilon) = \sigma^2\frac{1-2\tau(1-\tau)}{\tau^2(1-\tau)^2}.$$

Both theoretical and empirical results support the use of the ALD in the context of quantile regressions, even when the true distribution of the data is not ALD (Sriram et al., 2013).

[5] Both financial indicators are freely available on the European Central Bank Statistical Data Warehouse (CISS) and on the Banque de France Webstat (FCI), respectively.

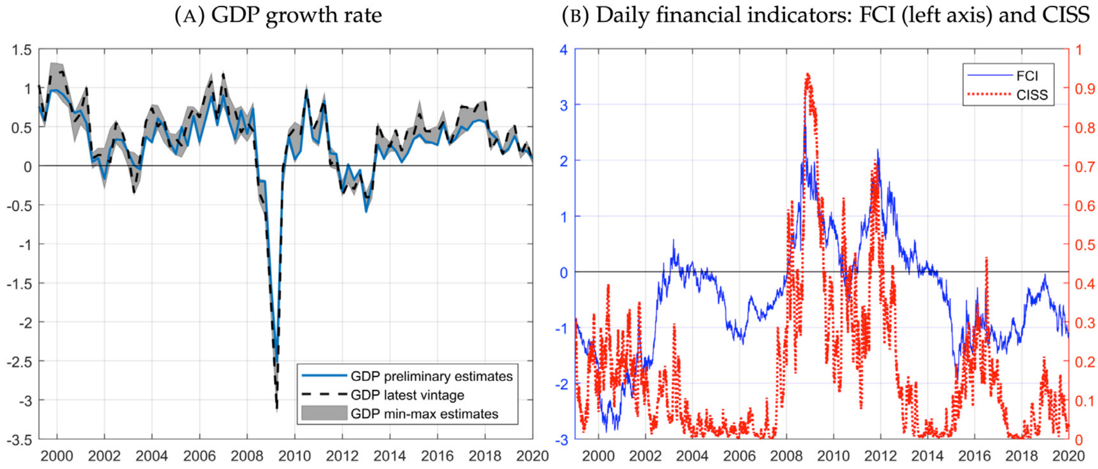

Fig. 2. Euro-area GDP and daily financial conditions indices (1999–2019). *Notes:* Panel A displays the GDP growth rate with the uncertainty associated with the estimates represented by the gray area. Panel B plots the two euro-area daily financial indicators: the financial conditions index (FCI) proposed by Petronevich and Sahuc (2019) and the composite indicator of systemic stress (CISS) proposed by Holló et al. (2012).

based on six main components (rates, credit, equity, uncertainty, inflation, and exchange rates) extracted from 18 daily series through principal component analysis. The FCI is then computed by aggregating these components using time-varying weights, which are based on univariate conditional volatilities estimated through a GARCH(1,1) model. As greater volatility increases the weight of the corresponding component, the FCI may put relatively more weight on single stressed components, whose signal is hence not muted by the state of the other components. As a result, the FCI resorts to information stemming from the actual level of the components and their volatility.

Finally, we control for macroeconomic news by setting $w_t^{(m)}$ as the Euro Area Composite Purchasing Managers Index Output (PMI hereafter).[6] This choice is dictated by the fact that the financial indicators considered are not explicitly designed to reflect real side developments and expectations, although they may indirectly capture such signals. Controlling for real factors can be important to gauge the real-time effect of financial indicators on real activity (see also Plagborg-Møller et al., 2020 and De Santis & Van der Veken, 2020). In this sense, survey data, such as PMI, provide timely and mostly unrevised macroeconomic news information that can be easily exploited in our mixed-frequency framework.

3.2. Computing the high-frequency GaR(10%)

As shown in Fig. 2(b), the FCI and the CISS present very similar high-frequency patterns, pointing to a strong correlation between these two series. Using the two indicators together in quantile regression (8) would then likely introduce multicollinearity in the model, leading to poor inference and predictive results. Hence, the first step for the construction of our GaR(10%) measure consists in estimating regression (8) through the Bayesian approach described in Section 2.3 and using two distinct sets of regressors $\mathbf{X}$. Both include one lag of GDP growth and the PMI, but only one financial indicator, i.e. either the FCI or the CISS. In the regressions we use a third-degree Almon lag polynomial ($p = 3$) and two end-point restrictions ($r = 2$) for both daily and monthly predictors, and we set the high-frequency lag windows to $C_x = 60$ (days) and $C_w = 6$ (months). We use standard diffuse priors for the model parameters $\boldsymbol{\Theta}$ and σ ($\Sigma_0 = 9 \times \mathbf{I}$ and $a_1 = b_1 = 0.01$, respectively), except for the autoregressive lag of GDP, whose prior mean and variance are set such that the probability mass is concentrated below unity (0.5 and 0.1, respectively). The Gibbs sampler runs for $N = 250,000$ iterations, with the first 50,000 used as the burn-in period, and every 10th draw is saved. The estimation sample spans from 1999Q1 to 2019Q4 and the daily forecasts start on July 1st, 2009.

Let us denote $\widehat{Q}^{(n)}_{i,y_{T|T-h_d}}(\tau|\mathbf{X}_i)$ as the nth posterior conditional quantile estimate of $y_{T|T-h_d}$ provided by the Markov chain Monte Carlo (MCMC) algorithm, for $n = 1, \ldots, N$. The subscript i denotes whether the underlying specification includes the FCI ($i = 1$) or the CISS ($i = 2$). The τ-th conditional quantile point estimate is then given by:

$$\widehat{Q}_{i,y_{T|T-h_d}}(\tau|\mathbf{X}_i) = \frac{1}{N}\sum_{n=1}^{N}\widehat{Q}^{(n)}_{i,y_{T|T-h_d}}(\tau|\mathbf{X}_i),$$

that is, the average predicted value from regression (8) for each quantile τ. In the present paper we consider $\tau \in [0.10, 0.25, 0.75, 0.90]$.[7] The generalized skewed Student's distribution is then fitted on $\widehat{Q}_{i,y_{T|T-h_d}}(\tau|\mathbf{X}_i)$, and we hence recover our **high-frequency daily GaR(10%)** indicator (see Eq. (6)):

$$Q^*_{i,y_{T|T-h_d}}(\tau = 0.10|\mathbf{X}_i). \tag{10}$$

[6] We thank an anonymous referee for this suggestion.

[7] It is worth noting that y_{T-1} is not available until its actual publication, usually around 30–45 days after the end of quarter $T-1$. To overcome this issue without affecting the real-time nature of the analysis, we use the EuroCoin indicator as a proxy for past GDP growth. This indicator is released by the end of quarter $T-1$, and real-time vintages are available on the CEPR website.

Finally, note that the Bayesian estimation provides a natural estimate of the standard error of the quantile function, as the conditional likelihood implies that the conditional quantile function is normally distributed. We can hence provide a measure of uncertainty surrounding the estimated daily GaR(10%) by computing its credible interval at some $(1-\alpha)$ level. In order to obtain asymptotically valid credible intervals, we implement the correction to the covariance matrix of the posterior chain proposed by Yang et al. (2016).

3.3. Combining growth-at-risk measures

In the previous step, we computed competing Gar(10%) measures, based on two alternative—although broadly complementary—financial indicators. However, for ease of analysis and interpretation of the results, it would be preferable from the policymaker's point of view to summarize the information stemming from these two indicators into a single GaR measure. For this purpose, we adopt a strategy based on density forecast combination (see, for instance, Hall & Mitchell, 2007 and Geweke & Amisano, 2011). Specifically, we implement the following algorithm:

(1) From Section 3.2, the individual smoothed conditional predictive quantile functions $Q^*_{i,y_{T|T-h_d}}(\tau|\mathbf{X}_i)$, for $\tau \in (0, 1)$, are computed sequentially (i.e. on a daily basis from July 1st, 2009 onwards) for each specification including either the FCI ($i = 1$) or the CISS ($i = 2$).

(2) The individual predictive quantile functions are then converted into density forecasts, and a real-time measure of their predictive performance is computed. We choose the quantile weighted probability score (QWPS; Gneiting & Ranjan, 2011), which provides a metric for the evaluation of the predictive ability of a model by emphasizing the (left) tail of the estimated density forecasts.[8]

(3) Combination weights $\omega_{i,T-h_d}$ are computed recursively (i.e. from the first to the last business day of each quarter) using a discounted QWPS combinations method, similar to the point forecast approach of Stock and Watson (2004) and Andreou et al. (2013):

$$\omega_{i,T-h_d} = \frac{w_{i,T-h_d}^{-\kappa}}{\sum_i w_{i,T-h_d}^{-\kappa}},$$

where

$$w_{i,T-h_d} = \sum_{j=T_0}^{T_\ell} \delta^{T_\ell - j} \mathrm{QWPS}_{i,j},$$

with $\kappa = 2$ and $\delta = 0.9$ (the discount factor), and where T_0 is the point at which the first prediction is computed, and T_ℓ is the point at which the most recent prediction can be evaluated in real time.[9]

(4) Finally, the combined conditional predictive quantile function is computed recursively as:

$$Q^*_{y_{T|T-h_d}}(\tau|\mathbf{X}) = \sum_i \omega_{i,T-h_d} \times Q^*_{i,y_{T|T-h_d}}(\tau|\mathbf{X}_i).$$

From the obtained combined quantile function, we recover the high-frequency (daily) *combined* GaR(10%) indicator $Q^*_{y_{T|T-h_d}}(\tau = 0.10|\mathbf{X})$. This daily indicator, along with its estimated 90% credible interval, as well as the evolution of the weights $\omega_{i,T-h_d}$, are presented in Fig. 3(b) (panels A and B, respectively) over the period from 2010Q3–2019Q4.[10]

4. Empirical results

This section presents four applications on the euro-area economy which illustrate the practical interest of using a daily GaR measure. In the first application, we evaluate its nowcasting properties when trying to track real-time GDP growth. A second application focuses on the real-time evolution of the indicator during the European sovereign debt crisis. The third one highlights its strong link with the main monetary policy decisions taken between 2013 and 2018. Finally, we evaluate the behavior of our high-frequency measure during the first six months of the Covid-19 pandemic.

4.1. Nowcasting GDP

As a first illustration, we evaluate the overall nowcasting performance of our BMIDAS-QR model, beyond the GaR measure. For this purpose, we consider the predictive densities obtained from the conditional predictive quantile function $Q^*_{y_{T|T-h_d}}(\tau|\mathbf{X})$. These densities are reported in Fig. 4, along with the preliminary estimates of quarterly GDP. A visual inspection of the figure suggests that the conditional distributions can track the actual

[8] The QWPS is a quantile-weighted version of the CRPS (Gneiting & Raftery, 2007), given by $\mathrm{QWPS}(f, y) = \int_0^1 \mathrm{QS}_\tau(F^{-1}(\tau), y)\nu(\tau)\mathrm{d}\tau$, where $\mathrm{QS}_\tau(F^{-1}(\tau), y)$, is the quantile score, and $\nu(\tau)$ is a weighting function. While $\nu(\tau) = 1$ leads to the standard CRPS, in our application we choose $\nu(\tau) = (1-\tau)^2$, which emphasizes the left tail of the predictive distribution and implicitly reflect an asymmetric loss function weighing more on lower quantiles. In practice, we compute the QWPS by approximating the integral above through numerical integration, $\mathrm{QWPS}(f, y) = \frac{1}{J-1}\sum_{j=1}^{J-1} \mathrm{QS}_{\tau_j}(F^{-1}(\tau_j), y)\nu(\tau_j)\mathrm{d}\tau$, where $\tau_j = j/J$ and $J =$ 1E+04. Note that the latter is reasonably lower than the number of draws from the smoothed predictive density, fixed here at 5E+05. Popular alternative metrics for combining density forecasts, such as the log score or the CRPS (see Geweke & Amisano, 2011 and Pettenuzzo et al., 2016, among others), nevertheless provided very similar results.

[9] Given the size of the evaluation sample, we do not compute specific weights for each forecast horizon h_d, but rather stack $\mathrm{QWPS}_{i,T-h_d}$ in a single vector. Further, because of the pseudo-real-time nature of the analysis, $T_\ell \neq T - h_d$. It follows that $\omega_{i,T-h_d} = \omega_{i,T-(h_d+1)}$, as long as a new point is available for evaluation. Finally, we use the first 97 daily predictions, from July 1st, 2009 to June 30th, 2010, to warm up the combination scheme. Over the period from July 1st, 2009 to November 12th, 2009, we initialize the combination weights by setting $\omega_{i,T-h_d} = 0.5$.

[10] The daily GaR(10%) series is obtained over time in a pseudo-real-time manner: every daily prediction corresponds to a specific h_d forecast horizon depending on the date of the prediction and the number of business days in the quarter. For instance, the GaR(10%) value for June 30th corresponds to the prediction obtained from a specification with $h_d = 0$, while the value for April 30th corresponds to the prediction obtained from a specification with $h_d = 40$ (on average).

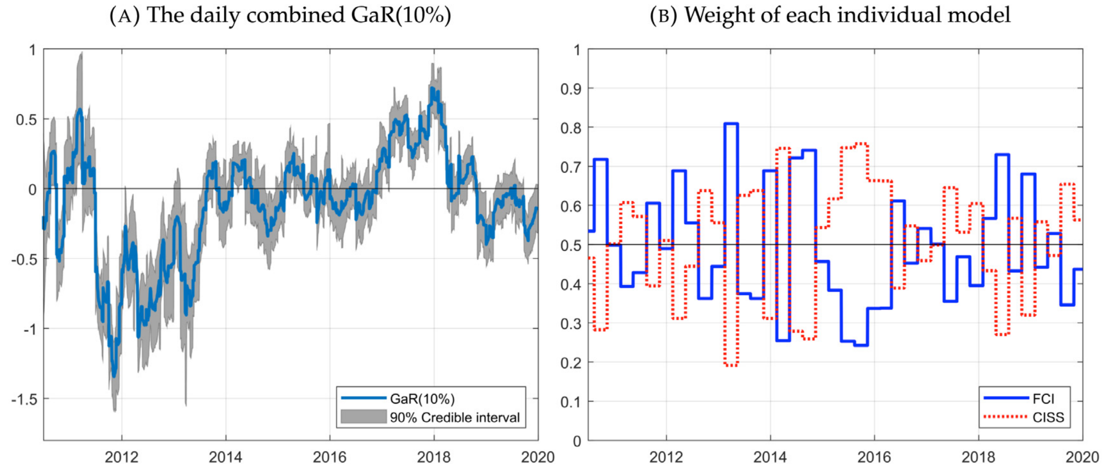

Fig. 3. The daily combined GaR(10%) and the weight associated with each individual model, over the 2010Q3–2019Q4 period. *Note:* The daily combined GaR(10%) corresponds to the lower 10th percentile of the distribution of expected real GDP growth, based on the combination of two individual GaR(10%) measures, estimated from models each including a different financial conditions indicator (FCI or CISS).

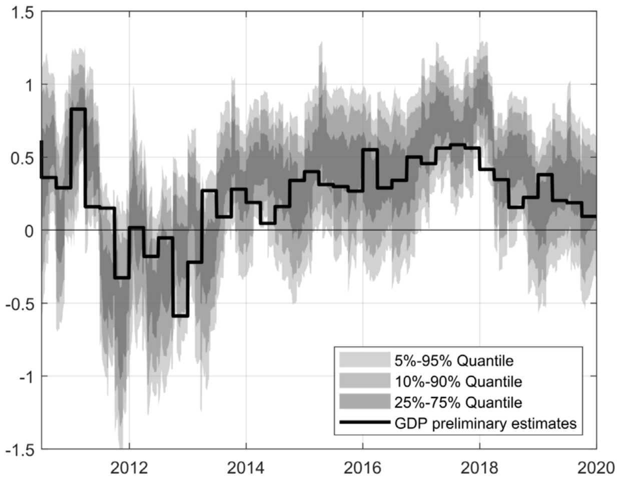

Fig. 4. Nowcasting GDP with the BMIDAS-QR model. *Note:* The BMIDAS-QR model is the Bayesian mixed-data sampling (MIDAS) model linking quarterly GDP and daily financial conditions indices estimated with a quantile regression (QR) methodology.

GDP growth fairly well, notably during the 2011–2013 recession episode (see Section 4.2), as well as during the acceleration of activity in 2016–2017 and the subsequent deceleration in 2018–2019.

Forecasts from our model are first compared to those from a benchmark specification, given by a simple Bayesian AR(1) regression (BAR). Consistent with the conditional distribution approach investigated in the present paper, we concentrate mainly on density forecasts, which are evaluated by means of four criteria: average log score differentials (LS), average continuous ranking probability score ratios (CRPS; Gneiting & Raftery, 2007), average quantile-weighted probability score ratios (QWPS; Gneiting & Ranjan, 2011), and average quantile score ratios (QS). For the CRPS, QWPS, and QS criteria, values less than one indicate that our combined model outperforms the benchmark. For the LS criterion, positive values indicate that our model produces more accurate density forecasts than the BAR. As a robustness check, we further consider density forecasts from two competing models, namely a combined Bayesian MIDAS model (BMIDAS) and Bayesian quantile AR(1) regression (BQAR). The combined BMIDAS density forecasts are computed using the combination strategy outlined in Section 3.3, where the underlying univariate models and densities are estimated using the approach described in Pettenuzzo et al. (2016). The BQAR model is similar to specification (8) but with only the lagged GDP in the quantile regression.

The results, reported in Table 1, point to the systematic outperformance of our BMIDAS-QR model with respect to the BAR(1) benchmark and the other competitors considered for relatively short forecast horizons (up to 20 business days). For longer horizons, this evidence is less clear-cut, as our model appears to outperform the competitors only according to the LS criterion, while the results mostly favor BQAR(1) regression. It is nevertheless worth noting that our model tends to outperform, although sometimes only slightly, the combined BMIDAS

Table 1
Out-of-sample results: Relative accuracy of density forecasts (2010Q3–2019Q4).

h_d	BMIDAS-QR				BMIDAS				BQAR(1)			
	LS	CRPS	QWPS	QS(0.10)	LS	CRPS	QWPS	QS(0.10)	LS	CRPS	QWPS	QS(0.10)
0	**0.30**	**0.81**	**0.81**	0.64	0.23	0.90	0.83	**0.63**	0.11	0.97	0.98	0.98
10	**0.22**	**0.91**	**0.86**	**0.60**	0.19	0.95	0.85	0.61	0.11	0.97	0.98	0.98
20	**0.26**	**0.88**	**0.84**	**0.60**	0.20	0.94	0.85	0.62	0.11	0.97	0.98	0.97
40	**0.09**	1.05	1.10	0.88	0.03	1.13	1.10	**0.84**	0.07	**0.99**	**1.01**	0.92
60	**0.11**	1.03	1.08	0.87	0.02	1.10	1.08	**0.82**	0.08	**0.99**	**1.00**	0.92

Notes: LS, CRPS, QWPS, and QS denote the log score, continuously ranked probability score, quantile-weighted probability score, and quantile score (evaluated at $\tau = 0.10$), respectively, in relative terms with respect to the BAR(1) benchmark. Bold values denote the best outcomes across the models considered for each forecast horizon h_d.

regressions, which embed exactly the same information as our quantile regressions.[11]

Overall, the empirical evidence reported here supports the importance of accounting for non-linearities when modeling and predicting real activity with financial indicators. Moreover, our results appear consistent with those for the United States reported by Plagborg-Møller et al. (2020), suggesting that financial data can provide accurate and timely indications of downward risks to GDP growth in the short term, while their contribution fades as macroeconomic information becomes progressively available.

4.2. European sovereign debt crisis

As a second illustration, we take the example of the European sovereign debt crisis that affected the euro area from 2010 to 2013. The 2008 financial crisis had indeed left its mark on public finances, leading to a significant increase in government bond yield spreads. Despite a rescue package for Greece, financial tensions intensified again due to the worsening of public finances in several other euro area countries and to the contagion arising from the undertaken agreement to restructure Greek sovereign debt by mid-July 2011. The sovereign debt crisis increasingly turned into a twin sovereign debt and banking crisis. Further negative feedback loops between vulnerable banks, indebted sovereigns, and weak economies took hold in several countries and led to acute financial fragmentation along country borders (Hartmann & Smets, 2018). Economic confidence fell, the economy slowed down rapidly, and the euro area entered a double-dip recession in the fourth quarter of 2011 until the first quarter of 2013, according to the CEPR business cycle chronology.

Fig. 5 shows that the GaR(10%) hovered between 0% and −0.5% during the first half of 2011. This is consistent with very mild risks to activity as, up to the end of June 2011, the growth rate of GDP for the euro area could have only be expected to run into slightly negative territory at 10% probability. As of mid-April 2011, the contagion effects of the deterioration in the sovereign CDS spreads started to signal that they may be long-lasting. The situation deteriorated in July 2011. On July 1st, the GaR measure started dropping rapidly, reaching −1.2% in the fourth quarter of 2011 when the euro zone entered into recession. This date precedes the announcement of the Moody's downgrade of Portugal on July 5th. This announcement, along with the continuing fears of a Greek default, could have triggered a sell-off in Spanish and Italian government bonds. By July 18th, the Italian government bond yields had increased by almost 100 basis points, while Spanish bond yields had increased by more than 80 basis points. The downgrading of sovereign ratings escalated and pushed bond rates up to critical levels for peripheral European countries. From April 1st, 2012, the GaR measure returned to its pre-crisis level, consistently with the first signals of an easing in financial conditions and an exit from the economic recession.

All in all, the results show that our daily GaR(10%) would have been able to correctly track this specific recession episode in real time. However, and more interestingly, the GaR(10%) measure tended to decline rapidly one quarter ahead of the beginning of the recession, dropping from a value close to zero to about -1% in only a few months. This swift change in the GaR can be interpreted as a possible early signal of recession led by a deterioration in financial conditions.

4.3. Announcements of unconventional monetary policy measures

We now turn to the link between the daily GaR measure and the unconventional monetary policy carried out by the ECB. Since 2013, the macroeconomic situation in the euro area has been characterized by increased risks threatening price stability and the anchoring of inflation expectations. Price developments have gradually moved away from values consistent with the ECB definition of price stability—i.e. a rate of inflation, as measured by the harmonized index of consumer prices (HICP), close to but below 2%. Faced with growing risks of disanchoring expectations, the Eurosystem responded by taking a number of unconventional measures. In particular, four major announcements were recorded between 2013 and 2018.

First, the ECB implemented forward guidance about the future course of monetary policy since July 4th, 2013. Forward guidance corresponds to a commitment on the future path of interest rates, so as to influence not only the short-term rates, which reached their lower bound close

[11] According to the tests proposed by Rossi and Sekhposyan (2019), we can reject the null hypothesis of correct calibration of the predictive densities of the BMIDAS-QR model at a 5% level only. In addition, the Diebold–Mariano–West test (Diebold & Mariano, 1995; West, 1996) for unconditional equal predictive accuracy (computed over point and density forecasts) tends to statistically support the outperformance of the BMIDAS-QR model over the BMIDAS model.

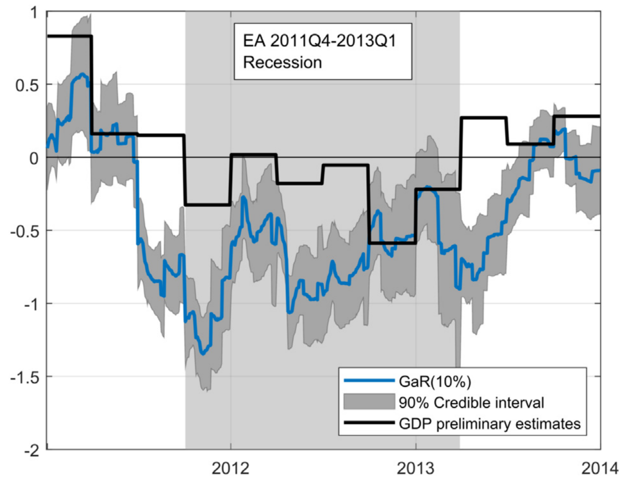

Fig. 5. GaR(10%) during the euro-area sovereign debt crisis. *Note:* The GaR(10%) corresponds to the lower 10th percentile of the distribution of the expected real GDP growth.

to zero, but mainly longer-term rates, which are to a large extent determined by expectations of future short-term rates. Second, the ECB decided to launch a large asset purchase program (APP) on January 22nd, 2015. It consisted of purchases on the secondary market of private securities and euro-denominated investment-grade securities issued by euro-area governments and institutions. The APP program was subsequently extended and adjusted on several occasions, notably by increasing both the duration and total amount of purchases. On March 16th, 2016, the ECB decided to extend the monthly purchases under the APP (extended APP) from 60 billion to 80 billion euros, including a new corporate securities purchase program (CSPP), starting in April 2016. This was intended to run until the end of March 2017, or beyond, if necessary. This measure was accompanied by a series of four targeted longer-term refinancing operations (TLTRO II) in order to ease private-sector credit conditions and to stimulate credit creation.[12] In addition, the rate on the deposit facility was lowered by 10 basis points to −0.40%. Finally, on December 8th, 2016, the ECB decided to adjust the parameters of the APP (adjusted APP): (i) the maturity range of the public-sector purchase program was broadened by decreasing the minimum remaining maturity for eligible securities from two years to one year, and (ii) purchases of securities under the APP with a yield to maturity below the interest rate on the ECB's deposit facility were permitted to the extent necessary. In addition, the ECB announced that the APP would continue at a monthly pace of 80 billion euros until the end of March 2017. From April 2017, the net asset purchases were intended to continue at a monthly pace of 60 billion euros until the end of December 2017, or beyond, if necessary.

In order to analyze the reaction of the daily GaR(10%) to the four monetary policy announcement dates, we performed a quasi-event study analysis, in the spirit of Fama et al. (1969). Event studies quantify an event's economic impact on so-called "abnormal returns". In our context, abnormal returns were calculated by comparing the observed GaR(10%) dynamics obtained in Section 3 with the GaR(10%) dynamics in the absence of the specific events ("normal returns"). To obtain these "normal returns", we chose an ARMA(1,1)-GARCH(1,1) process to efficiently capture the statistical properties associated with the mean and volatility of the actual GaR(10%) measure, and we estimated this model over a period starting in 1999 and ending 30 days before each specific announcement.[13] We then computed forecasts of the model over a 60-day period, centered on the event. The results of this exercise are provided on Fig. 6, in which the four monetary policy announcement dates are represented by vertical dotted lines. We observe that the daily GaR(10%) (plain line) largely deviates from the forecast path implied by the "normal time" representation during the 60-day event window. This result means that the GaR(10%) was very reactive to new monetary policy measures by increasing immediately after, sometimes before, each announcement, then exhibiting an upward trend in the following weeks. An exception may be the announcement of the APP, which had been anticipated by the markets for a while. The impact of the APP announcement on the GAR(10%) may only reflect a fraction of the overall financial market effect of all APP-related news. We also considered the case in which the GAR (10%) was purged from the effects of macroeconomic news, as the latter are likely to blur the interpretations. In this respect, we regressed the GAR(10%) on an intercept and dummy variables associated with the release dates of new GDP and PMI data and kept only the residuals and the intercept.[14] The results did not change, although the

12 The interest rate on these operations, each with a maturity of four years, was fixed at the main refinancing operations rate prevailing at the time of take-up.

13 The orders of the ARMA-GARCH process were selected according to standard information criteria (Akaike, Schwarz, and Hannan–Quinn).

14 The macroeconomic news concerning new numbers for GDP and PMI are the following: 01/07/2013 (GDP: Eurocoin 2013Q2), 20/06/2013 (PMI: 6th month), 01/01/2015 (GDP: Eurocoin 2014Q4),

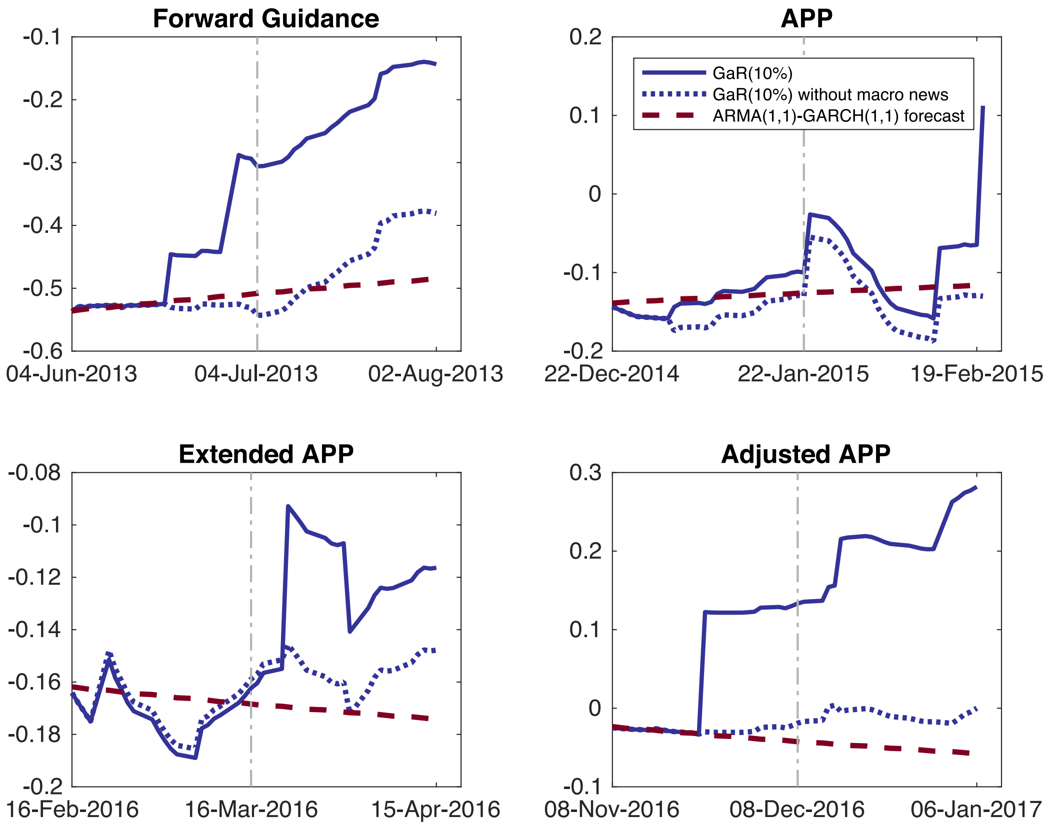

Fig. 6. GaR 10% and unconventional monetary policy announcements: An event study analysis. *Notes*: APP: asset purchase program, Extended APP: extension of the APP, and Adjusted APP: adjusted parameters of the APP. The vertical line corresponds to a specific announcement date. The GaR(10%) corresponds to the lower 10th percentile of the distribution of the expected real GDP growth (plain line). The dotted line corresponds to the implied GaR(10%) after including dummies with respect to macroeconomic news. The dashed line corresponds to the forecast path of an ARMA(1,1)-GARCH(1,1) representation of the GAR(10%).

amplitude of the deviations from the forecast path during the event window decreased (see the dotted line in Fig. 6). Consequently, from this quasi-event study analysis, we argue that this new high-frequency GaR(10%) measure can be useful for central bankers in order to check the immediate effects of their policies on macroeconomic risks and to subsequently adjust their monetary policy stance adequately.

4.4. Covid-19 pandemic

In this subsection, we focus on the Covid-19 crisis period that affected the euro area, as well as the global economy, during the first half of 2020. This shock was the most damaging event since the Great Depression and rather closer to a disaster, in Robert Barro's sense, than to a classical recession.

During the first two months of 2020, the impact of Covid-19 on the euro-area economy was basically insignificant, as it was not clear that the propagation of the coronavirus coming from China was about to turn into a global pandemic. The first anecdotal evidence came through the disruption of global value chains and diminishing external demand stemming from China. This lack of reaction from financial markets can be seen in Fig. 7: there is basically no shift in the conditional distribution of GDP growth for the first quarter of 2020 predicted by our model between January 24th (yellow curve) and February 26th (orange curve). Market sentiment started to turn negative in the last days of February, with the Euro Stoxx 50 dropping by about 27% by March 19th. Markets eased only after the ECB announced on March 18th the deployment of a new Pandemic Emergency Purchase Program, with an envelope of 750 billion euros until the end of the year, complementing an initial, smaller plan of 120 billion euros decided on March 12th.

In fact, as shown in Fig. 8, the first significant decline in the GaR(10%) coincides with the World Health Organization announcement recognizing Covid-19 as a global pandemic that came on March 11th. Mid-March also corresponds to the start of stringent lockdown measures within euro-area countries. Then, we observe a drop at the end of March, due to the steep decline in stock prices, as previously mentioned, and the release of the March PMI. This can be also seen in Fig. 7, where the

13/02/2015 (GDP: 2014Q4), 23/01/2015 (PMI 1st month), 20/02/2015 (PMI: 2nd month), 01/04/2016 (GDP: Eurocoin 2016Q1), 22/02/2016 (PMI 2nd month), 22/03/2016 (PMI: 3rd month), 02/01/2017 (GDP: Eurocoin 2016Q4), 23/11/2016 (PMI 11th month), and 15/12/2016 (PMI: 12th month).

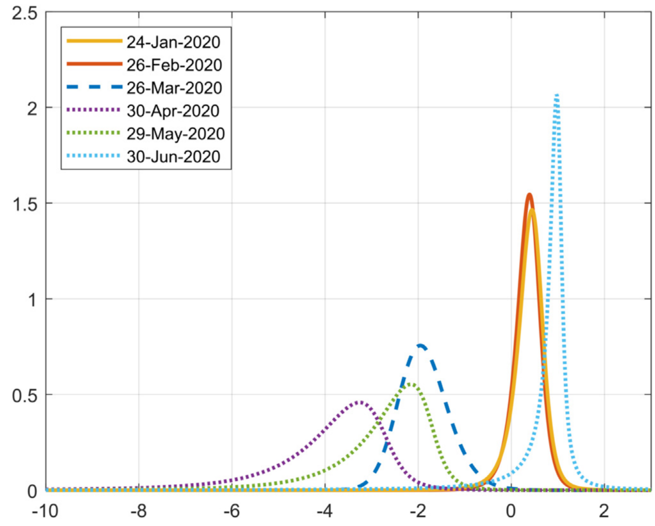

Fig. 7. Probability density functions of conditional GDP growth. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

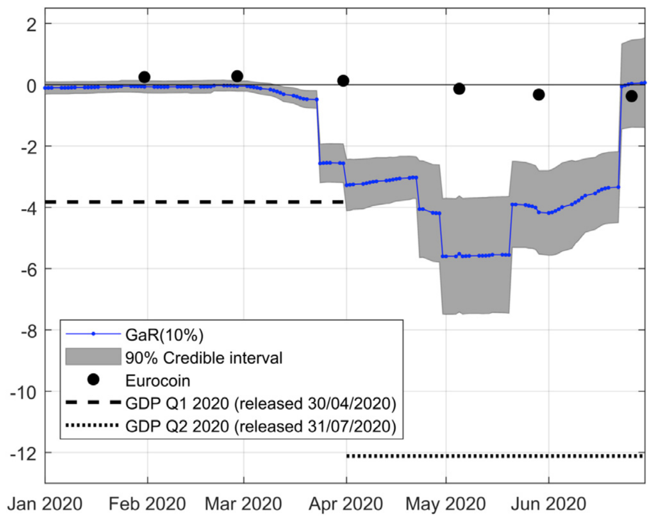

Fig. 8. GaR 10% and the Covid-19 pandemic. *Note*: The GaR(10%) corresponds to the lower 10th percentile of the distribution of expected real GDP growth.

predictive distribution of GDP growth for 2020Q1 estimated on March 26th began to sink into negative territory (dashed blue curve). Next, the daily GaR(10%) measure progressively falls to −6% at end of April, driven by both the PMI and the integration on April 30th of the preliminary estimate of GDP growth for 2020Q1 into the model. This shift at the end of April is also clearly visible in Fig. 7 (dotted purple curve). Starting from May 29th, the GaR(10%) started to increase again, driven by PMI appreciation, due to an improvement in business conditions and more optimistic expectations related to a sharp slowdown in the pandemic evolution before the summer of 2020.

Two interesting features arise from this empirical analysis. First, we compare our measure with the EuroCoin indicator provided by the CEPR, which can be interpreted as a real-time nowcast of euro-area GDP growth. As can be seen in Fig. 8, EuroCoin stayed quite high throughout the first half of 2020, showing only slightly negative values starting at the end of April onwards (−0.13), reaching −0.32 in May, while the strongly negative GDP growth in 2020Q1 was already known (−3.8% in quarter-on-quarter growth, published on April 30th). In 2020Q2, EuroCoin was also unable to reproduce the wide drop in quarterly GDP growth (−12.1%, as published on July 31st), only reaching a minimum of −0.64% in August 2020. Although it is fair to say that EuroCoin only aims at tracking a monthly *smoothed* estimate of quarterly GDP growth in the euro area, the deviation from actual GDP growth is nevertheless huge. A potential reason underlying this discrepancy is that this index is mainly based on macroeconomic information that comes with a delay,

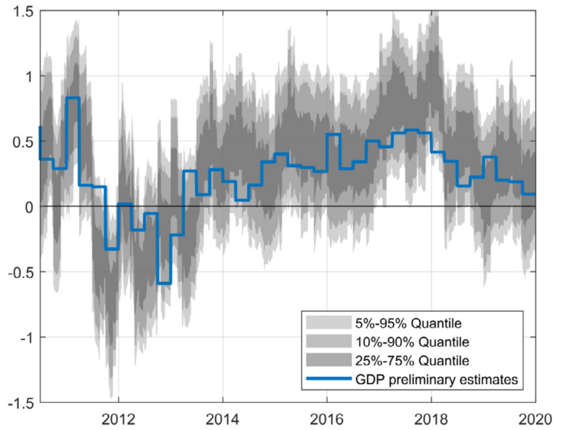

Fig. A.1. Nowcasting GDP with the combined BMIDAS regressions $y_t = \beta_1 y_{t-1} + \boldsymbol{\theta}_x' \widetilde{\mathbf{x}}^{(d)}_{i,t-h_d} + \boldsymbol{\theta}_w' \widetilde{\mathbf{w}}^{(m)}_{t-h_m} + \epsilon_t$. *Note:* $\widetilde{\mathbf{x}}^{(d)}_{i,t}$ denotes either the FCI ($i = 1$) or the CISS ($i = 2$). Predictive densities are combined using the QWPS metrics for the computation of combination weights.

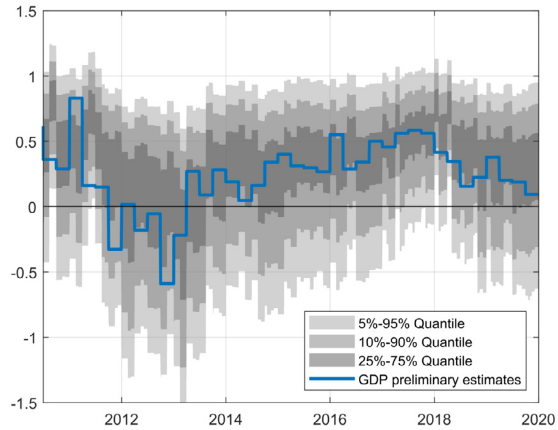

Fig. A.2. Nowcasting GDP with the BQAR(1) regression $y_t = \beta_1(\tau) y_{t-1} + \xi_1 \tilde{\nu}_t + \xi_2 \sqrt{\sigma \tilde{\nu}_t} \omega_t$.

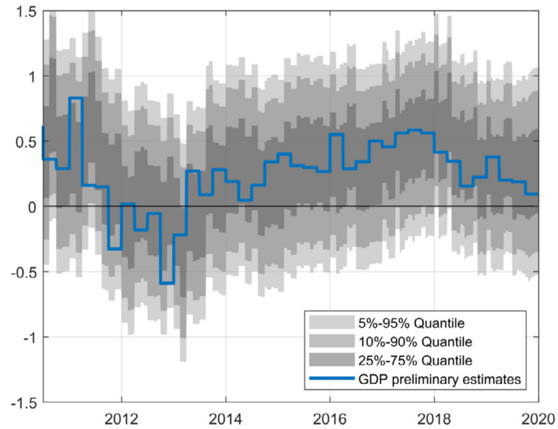

Fig. A.3. Nowcasting GDP with the BAR(1) regression $y_t = \beta_1 y_{t-1} + \epsilon_t$.

and that industrial production information is likely to be over-weighted in the index. By contrast, our daily GaR measure provided a clear signal of imminent deterioration in economic activity by mid-March.

Second, despite the timeliness of the signal provided by our daily GaR, its amplitude was lower than the drop observed in GDP growth. This gap can be explained by the nature of the shock underlying the Covid-19 recession. Indeed, it turns out that this recession can be understood as a mix of supply and demand shocks amplified by an uncertainty shock. On the other hand, the financial shock has been quite limited so far, mainly due to the swift and strong monetary policy response of the ECB. The synchronized reaction of the largest central banks around the world also likely contributed to globally sustaining the financial sector. In our view, this explains the limited shift and skewness of the estimated distributions presented in Fig. 7. However, the financial risk to GDP growth is still present, as we do not know how the current crisis will evolve in the upcoming months. In this respect, we think that our daily GaR measure will continue to be useful to track future risks to economic growth stemming from the financial sector during this major adverse economic event.

5. Conclusions

In this paper, we extended the quarterly growth-at-risk (GaR) approach of Adrian et al. (2019) by accounting for the high-frequency nature of financial conditions. Specifically, we used Bayesian mixed-data sampling (MIDAS) quantile regressions to exploit the information content of a financial stress indicator and a financial conditions index to construct real-time high-frequency GaR measures for the euro area. We showed that our daily GaR indicator: (i) displays good GDP nowcasting properties, (ii) can provide an early signal of GDP downturns, and (iii) allows for day-to-day assessments of the effects of monetary policies. During the first six months of the Covid-19 pandemic period, it provided a timely indication of tail risks on euro-area GDP. This new high-frequency GaR measure can be efficiently used by monetary policymakers to assess the impact of monetary policy decisions on macroeconomic risk through the lens of financial market perception.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Competing nowcasting models

See Figs. A.1–A.3

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